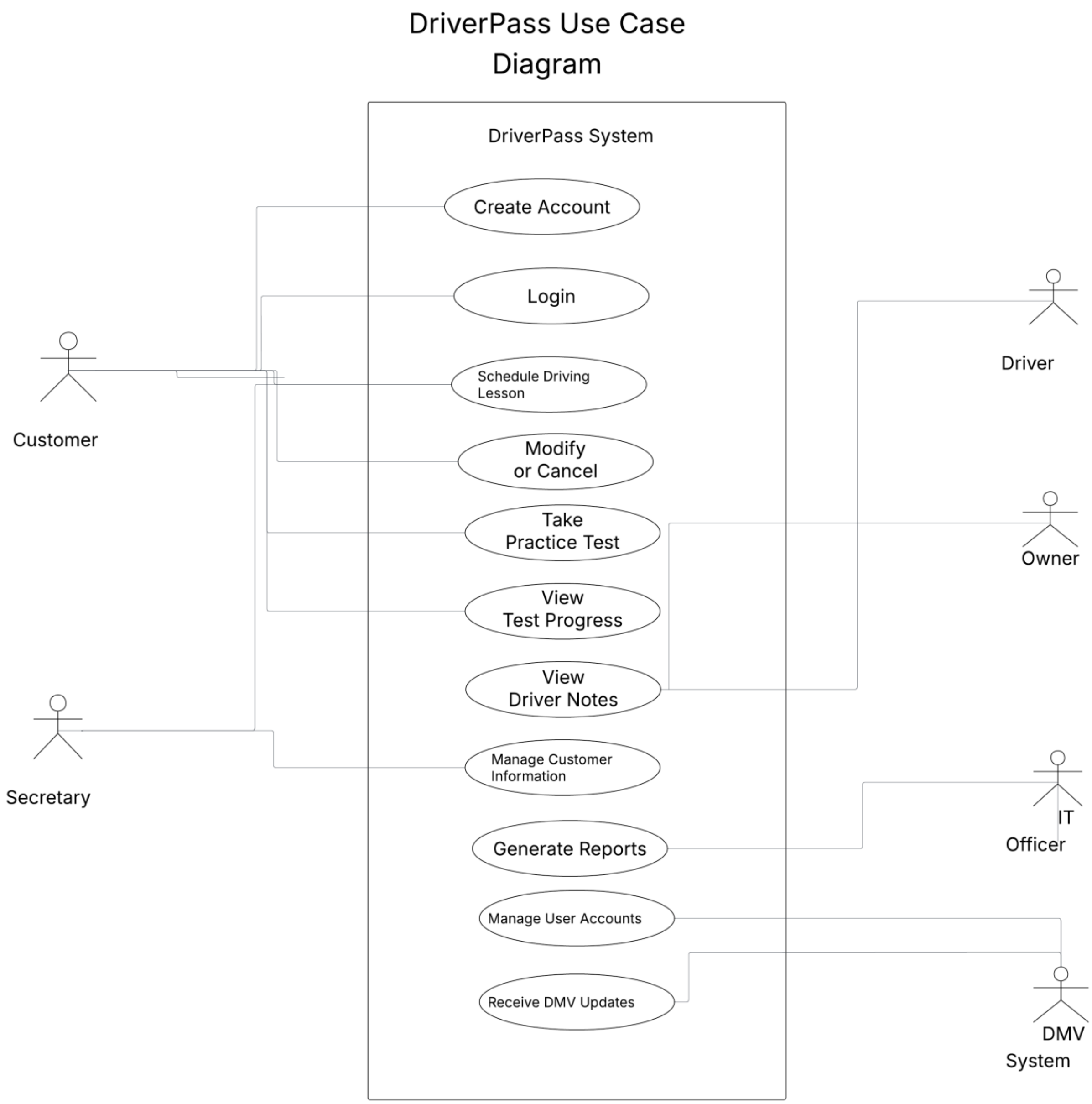
CS 255 System Design Document

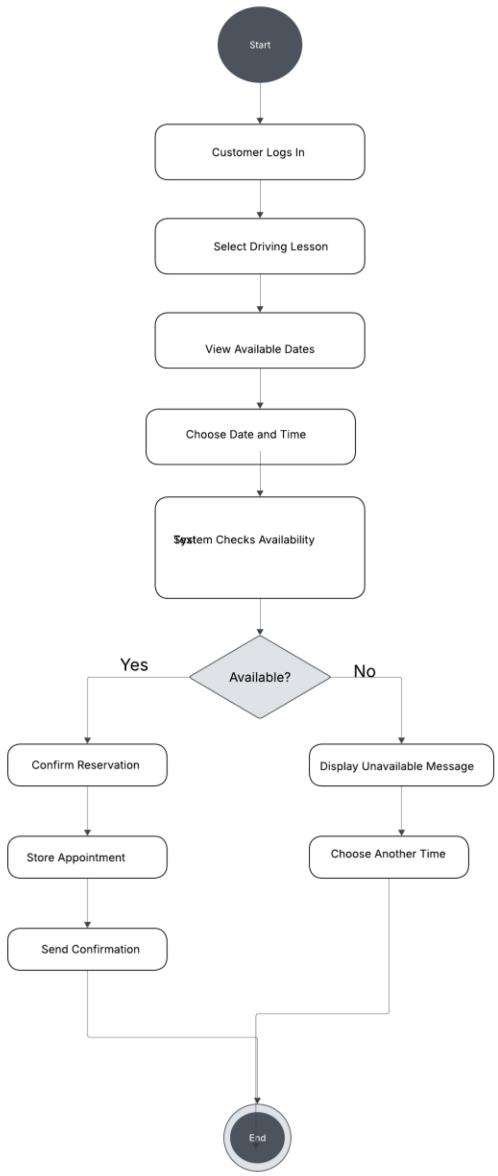
UML Diagrams

UML Use Case Diagram

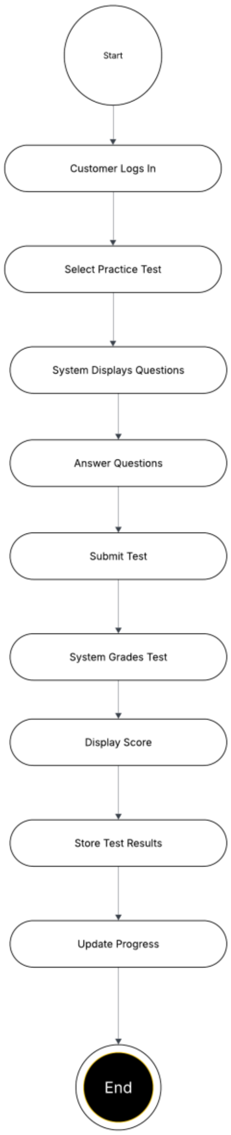


UML Activity Diagrams

Schedule Driving Lesson Activity Diagram

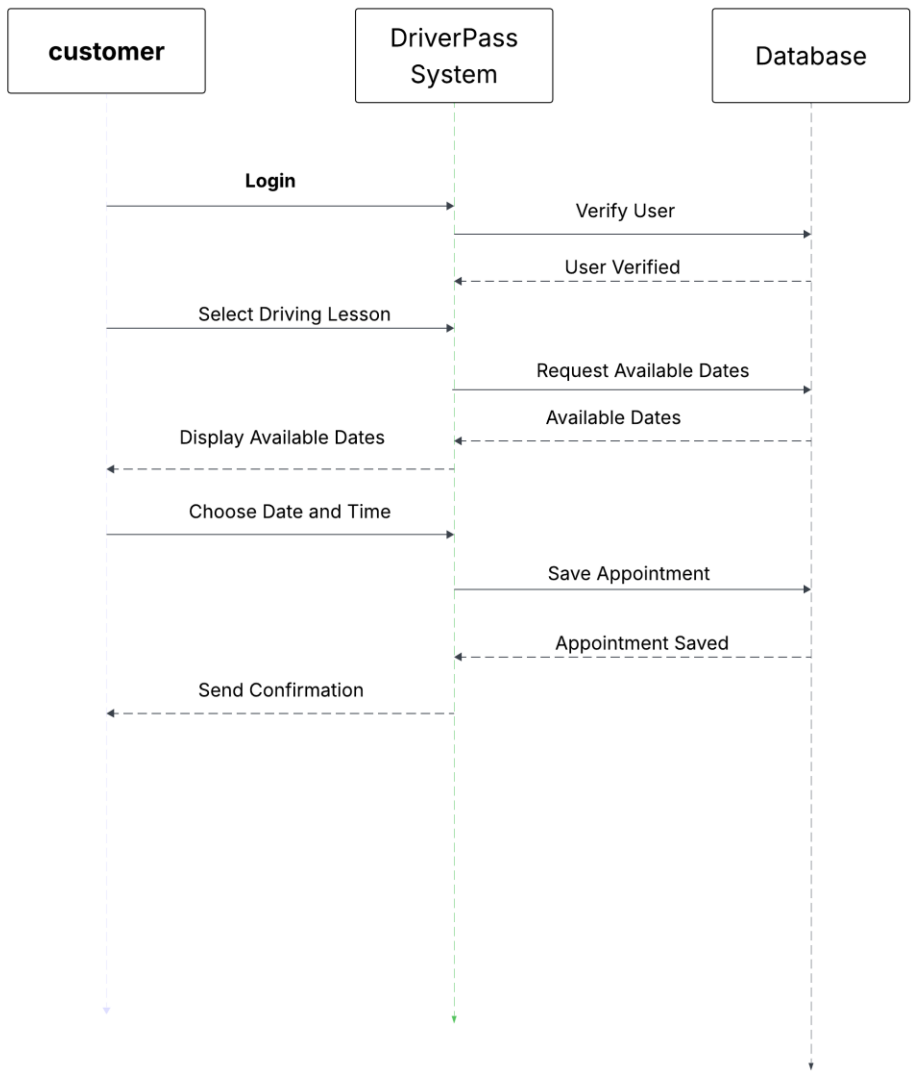


Take Practice Test Activity Diagram



UML Sequence Diagram

DriverPass Sequence Diagram – Schedule Driving Lesson



UML Class Diagram

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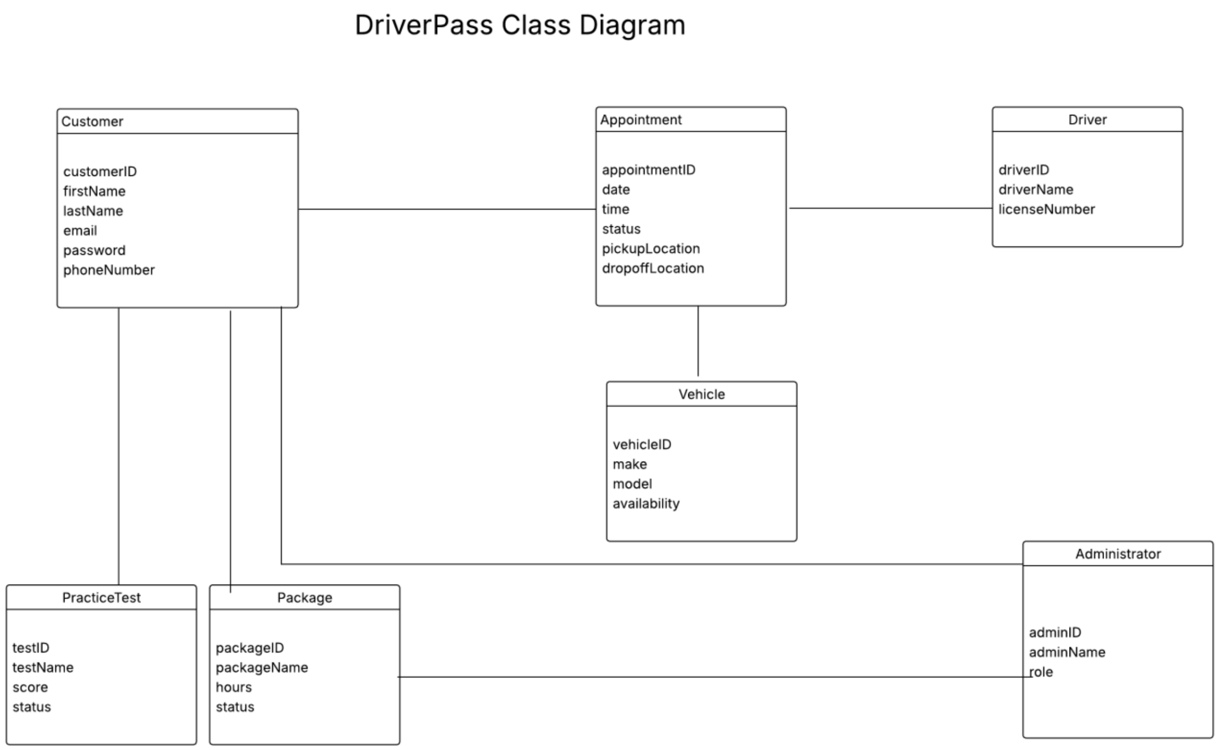

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Technical Requirements

The DriverPass system will be a web-based, cloud-hosted application that allows customers to take online practice tests, schedule driving lessons, manage appointments, and access DMV training materials. Users will access the system through a web browser on desktop computers, laptops, tablets, or mobile devices.

Hardware requirements include web servers to host the application, database servers to store customer and appointment information, backup storage systems, and reliable internet connectivity. Customers, instructors, administrators, and IT staff will need internet-connected devices to access the system.

Software requirements include a modern web browser, a secure database management system, and cloud-hosting services. The system must support user account management, appointment scheduling, online testing, reporting, and DMV content updates.

Infrastructure requirements include cloud-based hosting, secure database storage, backup and recovery services, and network security tools. The system should support multiple users simultaneously while maintaining reliable performance.

Security requirements include username and password authentication, password reset functionality, role-based access control, encrypted data transmission, account monitoring, and protection of customer information. The system should temporarily lock accounts or notify the IT administrator after repeated failed login attempts.

The system should be scalable and adaptable to support future growth, including additional users, instructors, vehicles, training packages, and DMV content updates without requiring major system redesign.